Vision-Based Continuous Drone Navigation in Wind Environments Using PPO with a Vision Transformer

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***Abstract*—We study indoor drone navigation as a real-world reinforcement learning problem: a quadrotor must reach a goal through a cluttered corridor while resisting wind and avoiding collisions. The task has continuous controls, partial observability from a front camera, delayed consequences, and safety-critical failures, making it a natural fit for RL. We formulate the problem as an MDP/POMDP and train a multimodal *proximal policy optimization* (PPO) actor-critic with a *Vision Transformer* (ViT) image encoder fused with low-level state features. The final system is evaluated against DDPG and SAC baselines and eight controlled ablations spanning reward components (per-step time cost, goal bonus, distance scale), curriculum learning, advantage normalization, image vs. state observations, ViT vs. CNN encoders, and domain randomization. On the final-stage evaluation, PPO+ViT reaches 100% success with 0% collision; DDPG/SAC fail under the same cluttered windy setting. Supplementary evaluations probe stronger wind, heavier sensor noise, composite corridor-layout perturbations, and longer corridors; results must be read together with simulator caveats (obstacle density is clipped to $[0, 1]$, and L-turn geometry is not instantiated—see Section VIII).**

***Index Terms*—Autonomous drones, reinforcement learning, vision transformer, PPO, continuous control, PyBullet, curriculum learning.**

Code & multimedia

GitHub – clone / inspect sources:
github.com/Lohith248/vision-drone-navigation-wind

Demo – qualitative rollout (MP4):
SharePoint demo folder (MP4)

I. Introduction

Autonomous drones are useful for inspection, search-and-rescue, inventory monitoring, and indoor delivery. These settings require flight through constrained spaces where GPS is unavailable, obstacles can move, and airflow disturbances can push the vehicle away from safe trajectories. Classical planning and control pipelines usually require explicit maps, reliable localization, and manually tuned controllers; however, visual navigation from raw images is partially observable and highly coupled with control.

Why reinforcement learning? RL directly optimizes sequential decision making: the drone can learn how short-term velocity choices affect future collision risk and goal progress. Continuous-control RL is especially appropriate because the action is not a discrete left/right command but a velocity vector (v_x, v_y, v_z). We choose PPO as the primary algorithm because clipped on-policy updates are stable in high-dimensional policy spaces [1], while off-policy actor-critic baselines (DDPG/SAC) provide meaningful comparisons for continuous control [2], [3].

Contributions.

- A PyBullet drone-corridor environment with RGB observations, state features, continuous velocity control, moving obstacles, sensor noise, and Ornstein–Uhlenbeck (OU) wind.
- A PPO+ViT multimodal actor-critic trained with a six-stage curriculum.
- A detailed reward design and ablation analysis explaining what each experimental change tests and how it affects learning.
- Quantitative comparisons, generalization tests, and a linked demo video for qualitative inspection.

II. Related Work / Literature Survey

Continuous control RL. DDPG learns a deterministic continuous policy using an actor, critic, target networks, and replay buffer [2]. It is sample efficient in some low-dimensional tasks, but can become unstable when critic errors are amplified through the actor. SAC improves off-policy control by maximizing both reward and entropy, which encourages exploration and robust stochastic policies [3]. PPO uses a clipped surrogate objective to limit destructive policy updates, trading sample efficiency for reliable optimization [1]. This stability is valuable when the policy includes a vision encoder.

Vision-based RL and transformers. CNNs are common image encoders in RL, but Vision Transformers split the image into patches and use self-attention to model long-range spatial relationships [4]. In corridor navigation, such global context can help distinguish open passage structure from nearby clutter. We compare ViT against a CNN encoder and a state-only policy to test whether vision and the encoder choice matter.

Curriculum learning. Curriculum learning trains agents on progressively harder versions of a task [5]. For drone navigation, beginning with wide obstacle-free corridors reduces early collision-dominated rollouts; later stages introduce clutter, moving obstacles, wind, and noise. This mirrors real robotics practice: learn coarse navigation first, then stress-test robustness.

III. Problem Formulation

We formulate the environment as a finite-horizon Markov Decision Process with partial observability from camera

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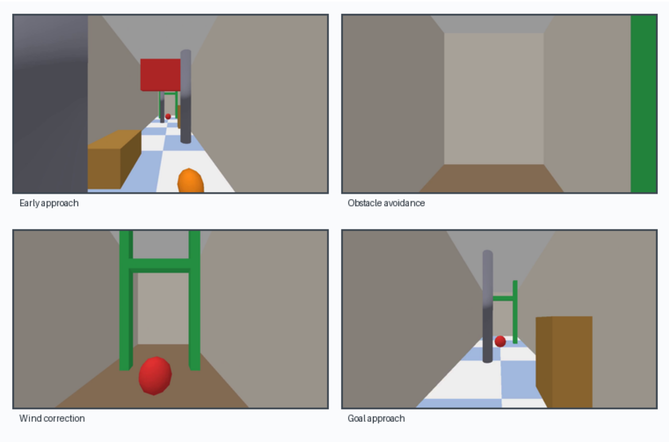

Fig. 1. Representative frames from the 720p demo video. The policy uses 64×64 observations internally; the figure is rendered at higher resolution only for visualization.

images:

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, \gamma, \rho_0, T).$$

State space. The true simulator state includes drone pose and velocity, obstacle poses, wall geometry, wind state, and the goal. The agent observes $o_t = [I_t, s_t]$, where $I_t \in \mathbb{R}^{64\times 64\times 3}$ is the front-facing RGB image and $s_t \in \mathbb{R}^{15}$ contains four wall distances, forward clearance, velocity, yaw error, goal direction, and the previous action. This makes the task a POMDP in practice.

Action space. The action is continuous:

$$a_t = (v_x, v_y, v_z) \in [-1, 1]^3,$$

which is scaled internally to ± 3.0 m/s and tracked by a velocity-PD controller.

Transition dynamics. PyBullet simulates rigid-body dynamics at 240 Hz; the policy acts at 30 Hz. OU wind is applied as an external force, producing temporally correlated disturbances:

$$dx_t = -\theta x_t \, dt + \sigma \, dW_t.$$

The transition also includes collision contacts, moving obstacles, sensor noise, and randomized corridor layouts from the environment seed.

Episode termination. Episodes terminate when the drone reaches the goal radius, collides with an obstacle/wall/floor, leaves valid bounds, or reaches the maximum step budget (1500 control steps).

Interpretation: The montage confirms qualitatively smooth progress along the windy cluttered corridor at demo resolution; observation preprocessing for control remains fixed at 64×64 RGB plus the 15-D state vector described in Section III.

IV. Methodology

A. Environment and Curriculum

The final training environment contains a 10 m corridor, static and moving obstacles, wind, and noisy sensors. To avoid early training being dominated by immediate collisions, we use the curriculum in Table I. The stages isolate different sources of difficulty: corridor narrowing tests lateral control; obstacle density tests perception and avoidance; moving obstacles test reactive control; wind and noise test robustness.

TABLE I
Curriculum stages used for PPO+ViT training.

Stage	Width	Obs. dens.	Moving	Wind σ	Noise
1. wide straight	3.0	0.0	no	0.00	0.00
2. narrow corridor	2.6	0.0	no	0.00	0.00
3. with obstacles	2.5	0.7	no	0.00	0.00
4. moving obstacles	2.5	1.0	yes	0.00	0.00
5. wind introduction	2.0	1.0	yes	0.25	0.05
6. full wind	2.0	1.0	yes	0.50	0.05

B. PPO+ViT Architecture

The policy is a Gaussian actor-critic. The image branch is a ViT with patch size 8, embedding dimension 128, depth 4, and 4 attention heads, producing a 192-dimensional visual feature. The state branch is an MLP mapping $15 \to 128$. The fused 320-dimensional representation is passed through a $256 \to 256$ shared trunk and split into an actor head $(\mu, \log \sigma)$ and critic head $V(s)$.

Interpretation: Dual-branch fusion lets proprioceptive rays and velocities stabilize low-level tracking while patch attention aggregates corridor-wide cues needed when clutter occludes local clearance readings.

C. Training Pipeline and Hyperparameters

The PPO rollout uses 4 vectorized environments, 2048 steps per rollout, 10 epochs, batch size 512, discount $\gamma = 0.99$, GAE $\lambda = 0.95$, clip range 0.2, entropy coefficient 0.01, value coefficient 0.5, gradient norm clipping at 0.5, and a linear learning-rate schedule from 10^{-4} to 10^{-5}. Mixed precision is enabled on GPU. Advantage normalization and clipped value loss are used to reduce update variance.

PPO+ViT was chosen as the primary method because the environment combines vision, continuous control, sparse catastrophic failures, and nonstationary curriculum stages. PPO is less sample efficient than replay-based methods but is more stable when the policy contains a deep visual encoder. DDPG and SAC were retained as baselines to test whether off-policy continuous-control methods can solve the same task with low-dimensional state observations.

V. Reward Design

The reward was designed to balance three competing goals: move toward the goal, avoid unsafe behavior, and prevent degenerate policies that survive without completing the corridor. The implemented report reward is

$$R_t = r_{\text{distance}} + r_{\text{goal}} + r_{\text{time}} + r_{\text{timeout}} + r_{\text{collision}}.$$

The distance term is normalized by the initial goal distance, so progress is comparable across layouts. Wind is not rewarded directly; instead, wind is introduced through transition dynamics. A good policy receives high reward only if it maintains progress

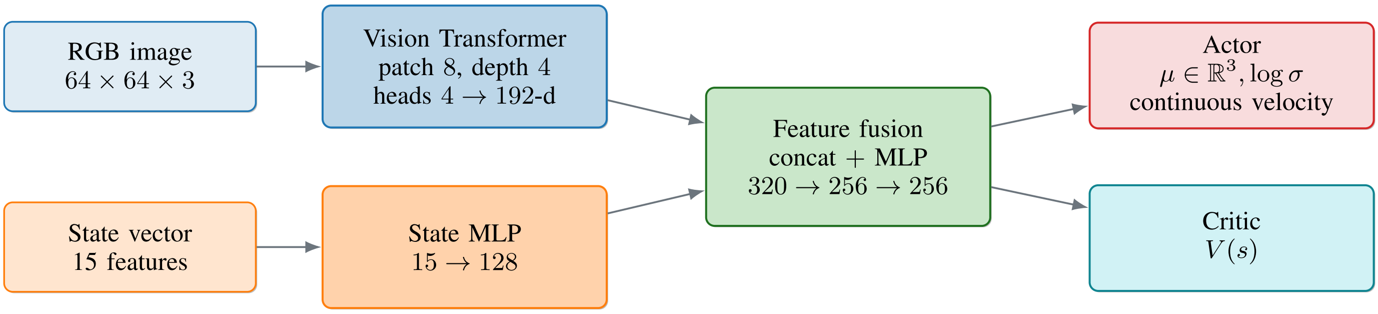

Fig. 2. Multimodal PPO actor-critic. The ViT provides global visual context while the low-dimensional state branch supplies geometric and control signals. The same fused representation supports both the policy and value estimate.

TABLE II
REWARD COMPONENTS AND INTENDED BEHAVIOR. VALUES ARE FROM THE FINAL REPORT CONFIGURATION.

Event / component	**Reward value**	**Purpose**	**Learning effect**
Forward distance progress	$100\,\Delta d/d_0$	Encourage movement toward the goal	Provides dense gradient before the sparse goal signal.
Goal reached	$+30$	Terminate successful episodes with high return	Separates true completion from merely surviving.
Collision	-50	Strongly discourage hitting walls/obstacles/floor	Makes unsafe exploration clearly suboptimal.
Timeout	-20	Discourage hovering or slow indecisive policies	Pushes the agent to finish within the horizon.
Per-step time cost	-0.002	Mild efficiency pressure	Breaks ties in favor of shorter, smoother paths.
Wind disturbance	implicit in $P(s_{t+1}\|s_t, a_t)$	Require robust correction under external force	Rewards policies that keep progress under perturbation.

despite wind disturbances. This avoids rewarding wind-specific artifacts and encourages robust closed-loop control.

Interpretation: Dense distance shaping supplies most gradient early; collisions and timeouts supply safety pressure; the sparse goal bonus and mild time tax primarily shape efficiency once flight is reliable.

the distance-scale ablation. Reducing the scaling factor from 100 to 1 preserves 100% success but increases episode length by nearly an order of magnitude with proportionally smaller shaped rewards, which strongly suggests the dense distance term is the dominant driver of sample-efficient progress, even though lighter shaping might still work given enough interaction time.

Component-removal ablations. To verify which reward terms actually drive learning, we retrained PPO+ViT twice from scratch with one knob set to zero in the reward configuration, keeping every other hyperparameter and the curriculum identical. Setting **time penalty** $-0.002 \to 0$ ($R_{NoTimePen}$) removes the implicit incentive to finish quickly; setting **goal bonus** $+30 \to 0$ ($R_{NoGoalBonus}$) removes the sparse terminal reward, leaving only the dense distance shaping and the collision/timeout penalties. Together with the existing *Low reward-scale* run, these three experiments isolate the role of each shaping term: dense-progress magnitude, sparse goal signal, and per-step efficiency pressure (Figure 3). At $N = 200$ episodes both component-removal variants still reach 100% success (Wilson 95% CI $[98.1, 100.0]$), but their efficiency profiles differ. Removing the time cost increases mean episode length from 150.5 to 252.1 steps with essentially unchanged return (124.75 vs 124.80), confirming the time penalty is an *efficiency tie-breaker*. Removing the $+30$ goal bonus drops mean return by almost exactly the missing terminal reward (94.37 vs 124.80) and increases mean length to 239.5 steps; the policy still finishes the corridor, but it loses the clean terminal anchor that previously

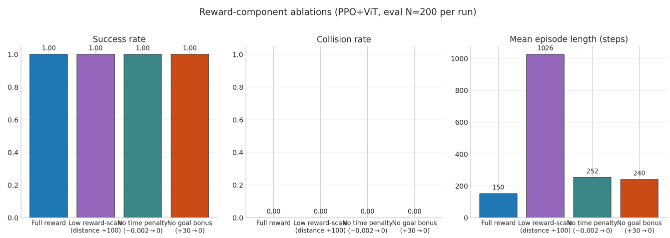

Fig. 3. Reward-component ablations. All four reward variants still solve the task at 100% success with 0% collisions; the difference is revealed by mean episode length. Removing the time penalty grows mean length from 150 → 252 steps, removing the goal bonus grows it to 240 steps with a ∼30-point return loss equal to the missing terminal bonus, and shrinking the distance scale by 100× pushes the policy to nearly the maximum step budget (1026 steps). This cleanly isolates the role of each shaping term in the design of R_t.

rewarded *committing to completion*, and it pays for that with longer, less decisive trajectories.

Interpretation: Success stays at 100% across variants; episode-length separation indicates dense shaping drives sample efficiency, goal bonus anchors terminal commitment, and time penalty trims path duration once collision risk is low.

VI. EXPERIMENTS AND ABLATION STUDIES

All evaluations use the latest checkpoint for each run and evaluate in the final curriculum stage with $N = 200$ episodes per run. Wilson 95% intervals therefore have a half-width of roughly seven percentage points at $p = 0.5$ (full width ≈ 14 percentage points). We ran experiments to answer five questions: (i) which RL algorithm is most stable; (ii) whether visual input is necessary; (iii) whether the ViT matters compared with CNN features; (iv) how curriculum, advantage normalization, and domain randomization affect the trained behavior; and (v) which individual *reward components* actually drive learning.

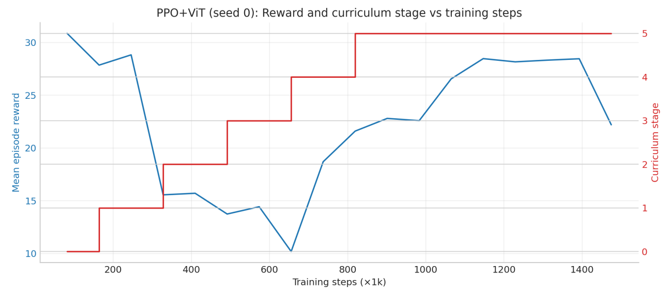

Fig. 4. Curriculum progression for the main PPO+ViT run. The policy first learns safe forward progress, then adapts to obstacles, moving obstacles, wind, and sensor noise.

Fig. 5. Training reward curves across methods and ablations. PPO variants show stable reward growth; off-policy baselines collapse early under the cluttered windy stage.

A. Algorithms Compared

PPO+ViT. Main method. It combines clipped on-policy updates with a multimodal visual policy. It is expected to be stable but less sample efficient.

DDPG. Chosen because the proposal required a Q-learning-family continuous-control baseline. It uses deterministic policy gradients and a replay buffer, which can be sample efficient but is sensitive to critic over-estimation.

SAC. Chosen as a stronger off-policy baseline. Entropy maximization should help exploration, but the method still depends on accurate value estimates.

Fair comparison reminder: Our "baseline" DDPG/SAC agents use state vectors without vision; Table IV contrasts that recipe against multimodal PPO+ViT under differing budgets and seeds.

B. Controlled Ablations

Table III lists the eight controlled variants relative to the main PPO+ViT run. Figures 4–5 show curriculum pacing and raw learning curves for context; Section VII then interprets final-stage scores.

Interpretation: Mean reward rises as stages advance; curriculum-stage staircase overlays show training dwelling progressively longer under harder dynamics without destabilizing early optimization.

Interpretation: Rolling mean rewards for multimodal PPO climbs steadily, whereas state-only DDPG/SAC traces saturate quickly once windy cluttered rollout dominates collision-heavy episodes.

VII. Results and Discussion

Interpretation: Bar heights summarize final-stage success, collision, and return under identical evaluation protocol; multimodal PPO+ViT dominates while CNN and off-policy baselines fail or regress once clutter and wind coincide.

Why multimodal PPO dominates these baseline snapshots. PPO's clipped objective stabilizes updates once collisions dominate variance; combined with fresh on-policy rollouts it avoids stale replay artifacts during curriculum shifts. *Crucially*, our "baseline" row pairs state-only observations with DDPG/SAC, whereas PPO+ViT consumes RGB plus proprioception—the observed gap reflects *both* algorithm and sensing modality. Matching algorithms fairly would require multimodal off-policy critics or state-only PPO with identical optimizer schedules.

Why DDPG and SAC collapsed. Both off-policy baselines rely on critic estimates. In this environment, small value errors are amplified because many actions lead quickly to collision, and the replay buffer mixes samples from changing curriculum stages. DDPG also has brittle deterministic exploration; SAC explores more, but still needs accurate Q-functions under high terminal penalties. Their short episode lengths and 100% collision rates show failure before meaningful goal-directed behavior emerges.

Why vision helped. State-only PPO receives wall distances and forward clearance but lacks rich spatial information about obstacle shape and visual layout. It still learns some navigation, but success drops to 89% (Wilson 95% CI $[83.9, 92.6]$) and the collision rate rises to 11% (CI $[7.4, 16.1]$); episodes also become much longer (319.8 steps vs 150.5 for the full PPO+ViT). That variant additionally uses a higher base learning rate (3×10^{-4} vs. 10^{-4} for the ViT recipe), so the gap mixes encoder capacity *and* optimizer tuning. Attention over image patches helps encode the global corridor opening and obstacle layout.

Role of curriculum. Training directly on the hardest curriculum stage still reaches 100% success in Table IV, albeit with $\approx 50\%$ longer episodes and modestly lower mean reward (124.52 vs. 124.80). Curriculum therefore accelerated convergence in our runs rather than unlocking behaviours entirely unattainable without staging.

Which reward terms drive learning. The two new reward-component ablations help separate *shaping* from *decision*. Removing the per-step time cost ($R_{\text{NoTimePen}}$) leaves dense distance progress, the goal bonus, and the safety penalties intact: the policy still reaches the goal with 100% success, but episode length grows from 150 to 252 steps, confirming the time penalty acts as an efficiency *tie-breaker*. Removing the $+30$ goal bonus ($R_{\text{NoGoalBonus}}$) is more subtle: the policy still completes the corridor at 100% success, but the headline reward drops by almost exactly the lost 30 terminal points (94.37 vs 124.80) and trajectories lengthen by 60%. Together with the low reward-scale run, this gives a three-point picture of how each reward term contributes: dense distance shaping is the dominant learning signal, the goal bonus anchors *commitment to*

TABLE III
NAMED TRAINING VARIANTS COMPARED AFTER CONVERGENCE AT THEIR RESPECTIVE LATEST CHECKPOINTS; HORIZONS, SEEDS, AND OCCASIONAL HYPERPARAMETERS (E.G. STATE-ONLY LEARNING RATE) DIFFER ACROSS ROWS. INTERPRET AS ILLUSTRATIVE SENSITIVITY SNAPSHOTS RATHER THAN STRICTLY MATCHED SINGLE-FACTOR ABLATIONS.

Experiment	**Change**	**Interpretation**	**Succ.**	**Coll.**	**Reward**
PPO+ViT full	ViT image + state, curriculum, adv-norm	Best overall; stable through full wind and clutter.	1.000	0.000	124.80
PPO seed 1	Same ViT recipe, shorter horizon	Alternative seed/checkpoint at 802k steps; longer episodes but same asymptotic success.	1.000	0.000	124.36
No adv-norm	Disable advantage normalization	Still succeeds; suggests reward scaling and curriculum already keep updates well conditioned.	1.000	0.000	124.24
Reward-component group					
Low reward-scale	Distance scale 100 $\to$ 1	Succeeds but with very long episodes; dense shaping mainly improves efficiency.	1.000	0.000	28.89
No time penalty	Per-step cost $-0.002 \to 0$	Negligible effect on success; verifies time penalty is a tie-breaker, not a driver.	1.000	0.000	124.75
No goal bonus	Goal reward $+30 \to 0$	Still 100% success, but mean reward drops by exactly the missing terminal anchor and episodes are 60% longer.	1.000	0.000	94.37
No curriculum	Train directly on stage 6	Still reaches 100% success here but with longer episodes and slightly lower mean reward versus the staged recipe (efficiency gap, not a missing-capability gap).	1.000	0.000	124.52
State-only PPO	Remove camera/ViT	Lower success and higher variance; ray/state features miss visual obstacle cues.	0.890	0.110	107.98
CNN encoder	Replace ViT with CNN	Fails on this setup; encoder choice strongly affects perception quality.	0.000	1.000	19.74
Domain rand.	Randomize mass, damping, wind, noise	Maintains success; useful for sim-to-real robustness.	1.000	0.000	124.55

TABLE IV
FINAL-STAGE COMPARISON USING $N = 200$ EVALUATION EPISODES PER RUN. SUCCESS AND COLLISION ARE REPORTED WITH WILSON 95% CONFIDENCE INTERVALS. REWARD AND EPISODE LENGTH ARE SAMPLE MEAN AND LENGTH OVER THE SAME 200 EPISODES.

Run	**Algo**	**Seed**	**Steps**	**Params**	**Success [95% CI]**	**Coll. [95% CI]**	**Reward**	**Ep. len**
PPO+ViT full	ppo	0	1,507,328	1,002,311	**1.000 [98.1, 100.0]**	**0.000 [0.0, 1.9]**	**124.80**	**150.5**
PPO no time-penalty	ppo	42	507,904	1,002,311	1.000 [98.1, 100.0]	0.000 [0.0, 1.9]	124.75	252.1
PPO+ViT + Domain Rand.	ppo	0	704,512	1,002,311	1.000 [98.1, 100.0]	0.000 [0.0, 1.9]	124.55	205.6
PPO no curriculum	ppo	42	507,904	1,002,311	1.000 [98.1, 100.0]	0.000 [0.0, 1.9]	124.52	227.7
PPO+ViT seed 1	ppo	1	802,816	1,002,311	1.000 [98.1, 100.0]	0.000 [0.0, 1.9]	124.36	279.2
PPO no adv-norm	ppo	42	507,904	1,002,311	1.000 [98.1, 100.0]	0.000 [0.0, 1.9]	124.24	224.9
PPO state-only	ppo	42	606,208	70,919	0.890 [83.9, 92.6]	0.110 [7.4, 16.1]	107.98	319.8
PPO low reward-scale	ppo	42	507,904	1,002,311	1.000 [98.1, 100.0]	0.000 [0.0, 1.9]	28.89	1026.0
PPO no goal-bonus	ppo	42	507,904	1,002,311	1.000 [98.1, 100.0]	0.000 [0.0, 1.9]	94.37	239.5
PPO+CNN	ppo	42	606,208	1,028,295	0.000 [0.0, 1.9]	1.000 [98.1, 100.0]	19.74	126.0
SAC state-only	sac	0	300,000	213,256	0.000 [0.0, 1.9]	1.000 [98.1, 100.0]	-41.97	12.2
DDPG state-only	ddpg	0	700,000	141,572	0.000 [0.0, 1.9]	1.000 [98.1, 100.0]	-53.76	19.7

finishing (so removing it preserves success but loses efficiency), and the time cost only trims the final paths. Removing all three at once would be expected to leave a policy that drifts forward without a strong reason to terminate.

VIII. GENERALIZATION ANALYSIS

Interpretation: Only the lengthened corridor reduces measured success; wind-noise stressors remain dark green because evaluation episodes still terminate successfully within $N = 100$ trials.

Interpretation: Bars mirror Table V: success and collision rates cluster near training saturation except for the 15 m corridor row where collisions dominate mid-flight before horizon exhaustion.

The policy maintains perfect measured success when wind torque doubles or sensor noise triples, reflecting robust closed-loop behaviour under those knobs. Two caveats prevent overstating layout robustness: (i) requested obstacle densities above 1.0 clamp to unity, so the "dense clutter" label currently

TABLE V
OOD GENERALIZATION, $N = 100$ EPISODES PER SCENARIO. SUCCESS AND COLLISION ARE REPORTED WITH WILSON 95% CIS.

Scenario	**Succ. [95% CI]**	**Coll. [95% CI]**	**Reward**	**Len.**
in-distribution	1.00 [96.3, 100.0]	0.00 [0.0, 3.7]	124.97	150.7
no wind	1.00 [96.3, 100.0]	0.00 [0.0, 3.7]	124.95	150.8
strong wind	1.00 [96.3, 100.0]	0.00 [0.0, 3.7]	124.94	150.8
dense clutter (ρ=1.5 $\to$ 1.0)	1.00 [96.3, 100.0]	0.00 [0.0, 3.7]	124.97	150.7
turns (flag only*)	1.00 [96.3, 100.0]	0.00 [0.0, 3.7]	124.72	149.3
narrow (composite†)	1.00 [96.3, 100.0]	0.00 [0.0, 3.7]	124.51	154.8
noisy sensors	1.00 [96.3, 100.0]	0.00 [0.0, 3.7]	124.66	156.6
long corridor (15 m)	0.19 [12.5, 27.8]	0.81 [72.2, 87.5]	28.76	135.5

*No extra corridor polygons yet wire through `enable_turns`; metrics mainly reflect mixed knob tweaks. †Narrow width paired with lighter clutter/wind versus training—still informative stress despite coupling.

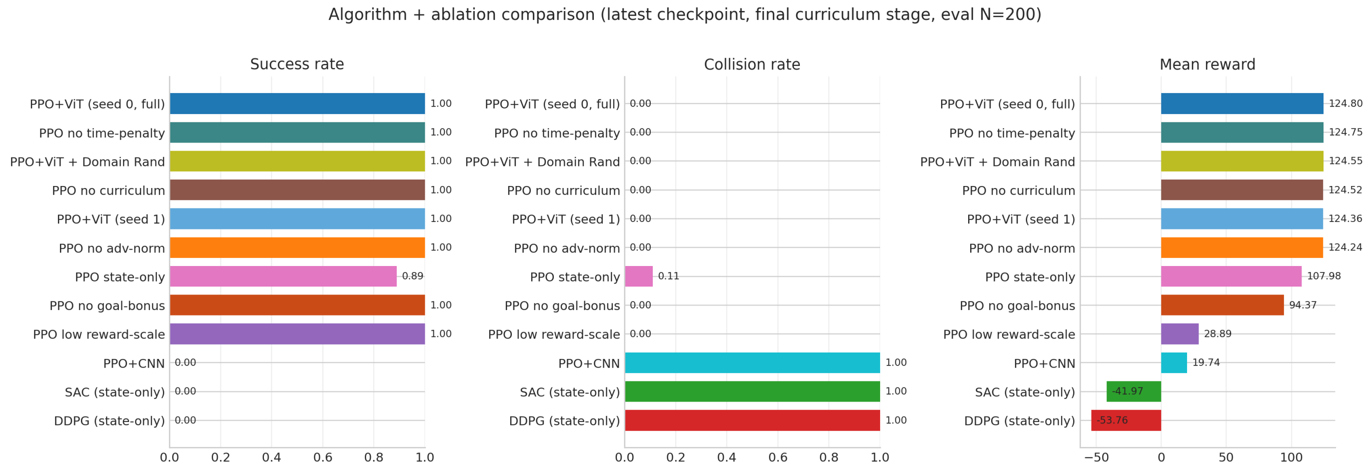

Fig. 6. Comparison across algorithms and ablations. Off-policy baselines are state-only; multimodal PPO drives the highest returns with nearly zero collisions under our evaluation protocol.

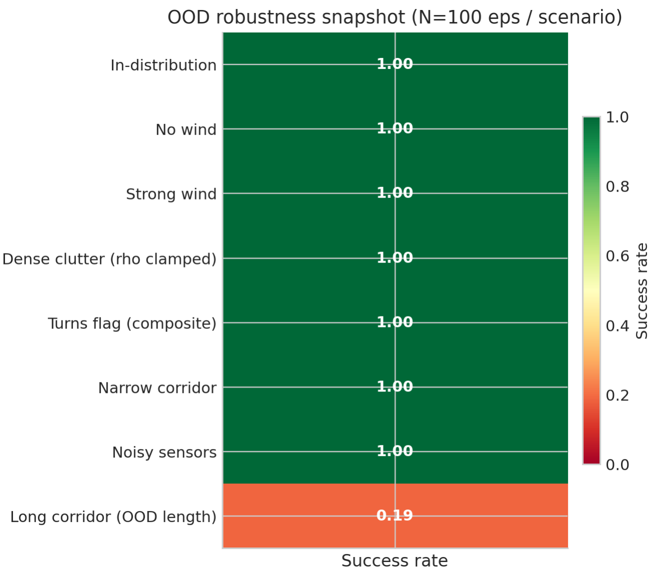

Fig. 7. Success-rate snapshot ($N = 100$). Apart from corridor-length extrapolation, successes persist across stronger wind and louder sensing noise. *dense clutter* repeats training-case obstacle saturation because densities beyond 1.0 clamp.

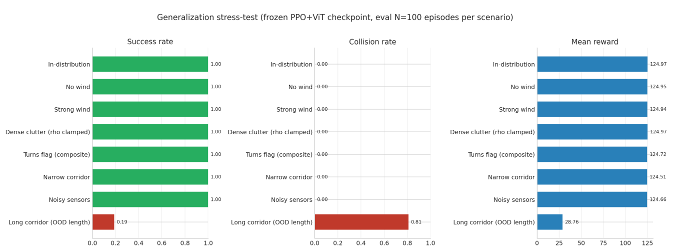

Fig. 8. Detailed stress-test bars. Failures concentrate on longer corridors. *turns* and *narrow corridor* rows sweep multiple dynamics knobs concurrently (*narrow corridor* also eases clutter density/wind relative to training). *dense clutter* matches the generalization summary table (Table V): densities > 1 clamp.

mirrors training saturation (duplicate metrics in the OOD table, Table V); (ii) the `enable_turns` flag does not yet reshape corridor meshes into true L-turn geometry—the "turns" scenario therefore combines nominal density/wind adjustments rather than encoding genuine bend topology. The "narrow corridor" scenario narrows width while simultaneously easing clutter density and wind relative to training, so it should be read as a composite perturbation rather than a pure width sweep. The stark failure mode remains corridor-length extrapolation ($15\,\mathrm{m}$ vs. $10\,\mathrm{m}$ training length): success drops to 19% with frequent mid-corridor collisions before episodes consume their expanded horizon budget. Future curricula should extend randomized corridor length (and finish geometry hooks for bends) before claiming clutter or layout extrapolation.

IX. Conclusion

This project demonstrates a complete RL pipeline for vision-based continuous drone navigation. The final PPO+ViT policy solves the full windy corridor task with 100% success and 0% collision in final-stage evaluation, whereas state-only DDPG/SAC runs collapse under the hardest windy cluttered stage under their budgets. The ablations show that the result is not just a final-score artifact: reward scale affects efficiency, vision improves reliability, the ViT encoder is substantially stronger than the CNN variant in this setting, and domain randomization preserves performance while preparing for sim-to-real transfer. The project therefore satisfies the core RL expectations: real-world motivation, formal MDP/POMDP definition, multiple RL methods, principled reward shaping, curriculum learning, and analytical comparison.

X. Limitations and Future Work

Limitations. First, the simulator is PyBullet rather than Flightmare/Isaac, so image realism is limited even though dynamics and collision contacts are modeled. Second, the policy is not deployed on a physical drone; real hardware would introduce latency, camera calibration, actuator limits, and safety constraints. Third, only two PPO seeds and one seed per ablation were feasible on the available single GPU; with more compute, every ablation could be repeated with multiple seeds and full Wilson CIs would be reported per row. Fourth, the long-corridor failure (19% success at 15 m corridor length) shows limited extrapolation outside the training geometry: the curriculum randomized width, clutter, wind, and noise, but kept corridor length fixed at 10 m. Fifth, scripted stress scenarios inherit simulator bookkeeping limits (obstacle density clipping and turn geometry wiring); Section VIII cautions how to interpret those rows until builders expose richer perturbations.

Future work. The most direct improvements are: (i) randomize corridor length, textures, lighting, mass, damping, wind correlation, and control latency; (ii) test recurrent policies (GRU/LSTM) for longer-horizon wind effects; (iii) improve exploration for off-policy baselines using TD3/SAC with visual encoders and targeted replay; (iv) use stronger transformer backbones or hybrid CNN-ViT encoders; (v) study multi-agent RL for dynamic obstacle negotiation; and (vi) transfer to a small real drone using domain randomization and a safety filter.

Reproducibility

Reproducibility commands

(1) Train baselines / selected ablations *(writes checkpoints under runs/)*

```
# Train queue helper (optional wrapper around
    configs)
./scripts/run_queue.sh

# Reward-only ablations from scratch (examples)
python -m drone_rl.train --config drone_rl/
    configs/ablations/ppo_no_time_penalty.yaml
    --no-resume
python -m drone_rl.train --config drone_rl/
    configs/ablations/ppo_no_goal_bonus.yaml --
    no-resume
```

(2) Evaluations & Wilson confidence intervals *(updates CSVs consumed by the tables)*

```
python scripts/aggregate_all.py --episodes 200
    --device cuda
python scripts/eval_generalization.py --
    checkpoint runs/best_vit/ppo_seed0_t1p5m/
    checkpoints/checkpoint_1507328.pt --config
    drone_rl/configs/ppo_best.yaml --episodes
    100 --device cuda --out report_artifacts/
    generalization.csv
python scripts/wilson_ci.py --in
    report_artifacts/all_runs.csv --episodes
    200 --out report_artifacts/all_runs_ci.csv
python scripts/wilson_ci.py --in
    report_artifacts/generalization.csv --
    episodes 100 --out report_artifacts/
    generalization_ci.csv
```

(3) Figures & `report.tex` auto-fill *(PNG plots + numeric injection)*

```
python scripts/figures/make_figures.py
python scripts/fill_report_tex.py
```

References

[1] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov. Proximal Policy Optimization Algorithms. arXiv:1707.06347, 2017.
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[3] T. Haarnoja, A. Zhou, P. Abbeel, and S. Levine. Soft Actor-Critic: Off-Policy Maximum Entropy Deep Reinforcement Learning with a Stochastic Actor. ICML, 2018.
[4] A. Dosovitskiy et al. An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. ICLR, 2021.
[5] Y. Bengio, J. Louradour, R. Collobert, and J. Weston. Curriculum Learning. ICML, 2009.